GENERAL BIOLOGY 102
VIRUS

Virus:
Virus is a small, simple infectious agent.
It can be defined as a non-cellular, infectious entity made up of genetic material and protein that can invade and reproduce only within the living cells of bacteria, plants and animals. The nucleic acid of viruses, which serves as genetic material, is **either DNA or RNA**(a small piece of genetic information ,DNA or RNA) inside a protective protein coating called **capsid**. Viral infections are any illness you get from virus and this virus uses the cells of living organisms to reproduce or make copies of themselves in a process called **replication**. Since human cells have the machinery needed to make more cells and virus do not, they have to break into cells of living organisms to be able to **replicate** and have higher load which makes its carrier sick. The Low risk viruses go away on their own, while the high risk viruses cause life-threatening or chronic illness. Viruses usually enter the body of its host through the mucous membranes(inner linings) of organs such as eyes, nose, mouth, stomach, penis, vagina and anus. Some viruses also get in through a break or a cut on the skin or from a bite from a mosquito or tick and other virus transmitting vectors. Viruses can cause several illness such as respiratory illnesses, sexually transmitted infections (STIs), skin conditions and many other kinds of illnesses. Some commonly known viral infections include: Common cold, The flu (influenza), COVID-19, Respiratory syncytial virus infection, Chickenpox, Measles, HIV/AIDS, Human Papillomavirus Virus(Warts on the body, genital etc), Herpes simplex (Herpes Simplex Virus), Poliomyelitis, Rabies, Mpox, Zika virus infection, Hepatitis etc.

Characteristics of Virus.
Virus is a sub-microscopic particle that lives by infecting the cells of a living organism. When compared with prokaryotic these are smaller in size and range from 20-300 nm. Individual virus is known as a **Virion**. Viruses do not have specialised machinery such as cell membranes, nucleus, mitochondria, endoplasmic reticulum, and many other cell organelles required for growth, reproduction, energy production, maintaining homeostasis etc. Viruses can replicate only by infecting a host cell. Viruses cannot be classified either a living organism or non-living organism. This is due to the fact that they possess certain defining characteristic features of living organisms and non-living entities. Viruses are significantly smaller than bacteria. They exist outside of the living cell as active particles surrounded by a protein coat.
Virus can replicate in plant, animal as well as bacteria cells etc. It enters and attaches itself to a specific host cell and injects its genetic material and reproduces by using the host genetic material and finally the host cell splits open, releasing the new viruses.

CLASSIFICATIONT OF VIRUS
Viruses are classified based on their different criteria;

1.CLASSIFICATION BASED ON THE PRESENCE OF NUCLEIC ACID
(A) DNA virus
The virus, having DNA as its genetic material. There are two different types of DNA virus
i. Single-stranded (ss) DNA virus are: e.g. Picornaviruses, Parvovirus, etc.
ii. Double-stranded (ds) DNA virus are: e.g. Adenovirus, Herpes virus, etc.
(B) RNA virus
The virus, having RNA as its genetic material. There are two different types of RNA virus

i. Double-stranded (ds) RNA virus are: e.g. Reovirus, etc.
ii. Single-stranded (ss) RNA virus are . It is further classified into two Positive sense RNA (+RNA) and negative sense RNA (-RNA). Poliovirus, Hepatitis A, Rabies virus, Influenza virus are examples of single-stranded RNA virus.

2. CLASSIFICATION BASED ON THE REPLICATION PROPERTIES AND SITE OF REPLICATION

Here, viruses invade into the host cell, where it replicates and assembly within the cell organelles.

(A).Replication within the cytoplasm of the host cell.
E.g. All RNA viruses except the Influenza virus.

(B). Replication within the nucleus and the cytoplasm of the host cell.
E.g. Influenza virus, Poxvirus, etc.

(C). Replication within the nucleus of the host cell.
All DNA viruses except Pox virus.

(D). Replication of the virus through the double-stranded DNA intermediate.
E.g. All DNA viruses, Retrovirus and some tumour causing RNA virus.

(E). Replication of the virus through a single-stranded RNA intermediate.
E.g. All RNA viruses except Reovirus and tumour-causing RNA viruses.

3.CLASSIFICATION BASED ON THE HOST RANGE

Based on the type of host, there are four different types of viruses:

(A). Animal viruses

These viruses infect by invading the cells of animals, including humans. Prominent examples of animal viruses include the influenza virus, mumps virus, rabies virus, poliovirus, Herpes virus, etc.

(B). Plant viruses

These viruses infect plants by invading the plant cells. Replication of plant viruses is obligate and does not happen without a host. Well-known examples of plant virus include the potato virus, tobacco mosaic virus (TMV), beet yellow virus, and turnip yellow virus, cauliflower mosaic virus, etc.

4.CLASSIFICATION BASED ON THE MODE OF TRANSMISSION

Airborne infections - This is the transmission of the virus through the air into the respiratory tract. E.g. Swine flu, and Rhinovirus.

Fecal oral route -This is the transmission of the virus through the contaminated water or food. E.g. Hepatitis A virus, Poliovirus, Rotavirus.

Sexually transmitted diseases- Transmission of the virus through sexual contacts with the infected person. E.g. Retrovirus, human papillomavirus, etc.

Transfusion-transmitted infections- Transmission of the virus through blood transfusion. E.g. Hepatitis B virus, Human Immunodeficiency Virus, etc.

Zoonoses -Transmission of the virus through the biting of infected animals, birds, and insects to human. E.g. Rabies virus, Alpha virus, Flavivirus, Ebola virus(emerging zoonotic disease) etc

TYPES OF VIRUS

Viruses are categorized into family and genus based on related features, **such as size, shape and the type of genetic material they carry**. Some common types of viruses include:

Influenza viruses (Orthomyxoviridae): The **Orthomyxoviridae** family of viruses includes influenza A and B, which cause the flu. Strains of influenza A also causes avian flu (bird flu) and swine flu (H1N1).

Human herpesvirus (Herpesviridae):
Herpesviridae is a large family of viruses. It have two variants 1 and 2. They cause several types of illnesses, like oral and genital herpes, chickenpox, shingles, Epstein-Barr and cytomegalovirus (CMV).
Coronaviruses:
These are a large family of respiratory viruses that includes Severe Acute Respiratory Syndrome COV-1(2002-2004) and Severe Acute Respiratory Syndrome COV-2, Middle east respiratory syndrome (MERS), Named due to their crown-like spikes on their surface. They have symptoms such as fever, cough, sore-throat, runny nose, and can progress to more serious ailments such as pneumonia or bronchitis.
Human papillomavirus (HPV):
Human papillomaviruses are part of the Papillomaviridae family of viruses. They are small, non-enveloped DNA virus that infects the skin or mucosal cells. This causes wart and some types of cancer.
Enteroviruses:
Enterovirus is a genus (one level smaller than the group called a family) of viruses that infect your intestinal tract. Types of enteroviruses cause polio virus disease, hand, foot and mouth disease.
Flaviviruses:
Viruses in this genus are often spread by **mosquitoes**. They cause illnesses like Zika virus infection(by **Aedes mosquitoes),** West Nile (by **Culex mosquitoes),** dengue fever (by **Aedes mosquitoes)** and yellow fever (by **Aedes mosquitoes)**.
Orthopoxviruses:
Viruses in the genus Orthopoxvirus causes flu-like symptoms and blistering rashes. examples are **Mpox, Cowpox .**
Hepatitis viruses:
Though they don't all belong to the same family or genus, hepatitis viruses all infect your liver leading to inflammation but they differ in modes of transmission, severity of the illness and prevention methods. There are **Hepatitis A,B,C,D,E**. **Oncoviruses:**
Oncoviruses are viruses that can cause cancer. Viruses that have been linked to specific cancers include: Human Pappilloma Virus, Human herpes virus 8 (HHV-8).
TYPES OF VIRAL INFECTION
Respiratory infections: Respiratory infections affect the nose, throat, lungs and airways. Some respiratory viruses can cause bronchitis, Sinusitis, Ear infections, pneumonia. Examples include Common cold, The flu(Influenza), Covid'19 etc.
Digestive systems infection: Viral infections in your digestive system can affect the stomach, and intestines or the liver. They can cause stomach flu(gastroenteritis).Examples: Norovirus, Rotavirus ,they can also cause chronic liver disease.egs. Hepatitis virus.
Viral hemorrhagic fever: These viral infection can affect how blood clots and can weaken the blood vessels, causing potentially life threatening bleedings. examples: Ebola(mainly fruit bats), yellow fever, Lassa fever.
Sexually transmitted infections: These are diseases mainly spread by sexual contact(anal, oral, vaginal). Examples: **HIV, Human Papilloma Virus, Hepatitis B, Herpes simplex virus**.
Exanthematous infections: These viral infections cause rashes that can appear as bumps, or rashes on the skin or spots of blood under the skin. They can also cause respiratory or other symptoms. Examples: Measles, Chicken pox, Mpox and other related infections.

Neurological infections: These viruses attack the cells of the nervous system (brain and spinal cord).It can cause paralysis, swelling (inflammation) of the brain or brain covering. Examples: **Polio, Rabies** etc.

Congenital infections: These are viral infections that occur from birth(present from birth). The pregnant woman passes it to her foetus while it is developing or during child birth. It can cause vision or hearing impairment, delays in development, and neurological conditions. Examples: Zika virus, Rubella(German measeles).

DISEASES CAUSED BY VIRUS

Common cold:

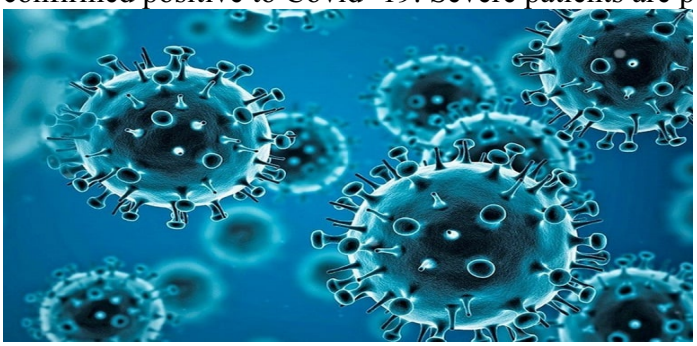
This is a viral infection that affects the upper respiratory tract, it can be caused by a variety of viruses but the most common cause is the rhinoviruses and it is transmitted from person to person or surface to surface. It presents gradual nasal symptoms such as stuffy nose, sneezing, loss of smell or taste, congestion, watery nasal secretions, sore throat, cough, fatigue, chills e.t.c.

The flu (influenza):

This is a respiratory ailment, an infection of the nose, throat and lungs. The causative organism of flu is influenza virus and It presents sudden symptoms such as fever, muscle aches, chills, sore throat, headache, shortness of breath, Dry persistent cough etc. Also dehydration, seizures, in children. Certain group of people are at higher risk of having flu complications than the others - Young children below one year, pregnant women, adults older than 65 years old, those taking care of the sick especially in the hospital, generally ,those with weakened immune system.

COVID-19:

Cooronavirus disease(Covid-19) is an infectious disease and are a subfamily of viruses. Covid-19 is caused by **SARS-CoV-2** (Severe Acute Respiratory Syndrome Corona Virus 2).They are crown like appearance. It spreads between people in close contact presenting common symptoms such as fever, chills, sore throat, persistent cough, shortness of breath, appetite loss, loss of taste and smell, difficulty sleeping etc. People who have preexisting medical conditions or those with compromised immune system are at higher risk of mortality when they have Covid-19. Examples are those with chronic heart, lungs or liver problems, HIV, diabetes, cancer or dementia patients etc. There are numerous measures towards management of Covid-19 infections, There is vaccine for Covid-19 which provide strong protection against serious illness, hospitalisation and death, Crowded places should be avoided, properly fitted mask should be worn, hands should be cleaned regularly with alcohol swab(or hand sanitizer with at least 60% alchohol), regular washing of hands with soap and clean running water as well as self isolation if confirmed positive to Covid -19. Severe patients are put under ventilators to prevent death.



Covid -19 virus

Respiratory syncytial virus :

This is a common respiratory virus that infects the nose, throat, lungs and breathing passages. usually causes mild, cold-like symptoms. RSV can cause serious illness in mostly infants, older

adults and people with compromised immune systems. It spreads through contact with respiratory droplets(sneezing, coughing or kissing) from an infected person or touching surfaces contaminated with the virus and then touching your eyes, nose, mouth. And has symptoms such as congested or runny nose, dry cough, sore throat, sneezing etc RSV is a vaccine preventable disease.

Chickenpox:

This is a highly contagious viral infection causing an itchy rash with small fluid-filled blisters on the skin. It spreads through. This disease mostly affects those who haven't had the disease or who haven't been vaccinated against it. The causative organism of Chicken pox is **Varicella - zoster** virus and it is transmitted from person to person through bodily fluids(coughing and sneezing, secretions from eyes, nose, mouth) and bodily contact and presents symptoms such as low grade fever, headache, an itchy blister-like skin rash, scabs after the blister breaks.

Measles (Rubeola):

This is an extremely contagious airborne disease causative organism of measles is an RNA virus called morbillivirus. It presents symptoms that usually appear 7-14 days after exposure, and it includes a fever (possibly up to 40°C), cough, runny nose, sneezing, watery eyes, body aches, white spots in the mouth and a red rash appearing about 3-5 days after the onset of the symptoms. This usually appears on the face, spreads to the chest and back. It is transmitted from person to person when an infected person breathes, coughs, sneezes, or talks and droplets from these body parts are inhaled by person who doesn't have the virus. Vaccines offer effective protection against measles.

HIV/AIDS:

The causative organism of AIDS is a retrovirus **Human immunodeficiency virus(HIV)** and it is transmitted from person to person through sexual contact (unprotected sex, anal sex, oral sex), mother to child during pregnancy and delivery, breast milk, blood transfusion with an infected blood, exchange of body fluids using unsterilized or contaminated needles. This viral disease attacks the immune system by destroying the CD 4 Cells (CD4 T Lymphocytes) of the immune system thereby exposing the infected person to other deadly infections and diseases such as tuberculosis, other sexually transmitted infections and cancer as a result of the immune system being compromised by the HIV infection. (It cannot be transmitted or spread through saliva, sweat, tears, sharing towels and beddings, insect vectors, toilet seats .etc. But can be present in semen, breast milk, blood, vaginal secretions etc. They also suffer from stigmatization from the society as a result of HIV infection, this can lead to depression or even death. Though there is no cure for HIV/AIDS, the antiretroviral drugs are administered to drastically reduce the viral load on the patient.

Human Papillomavirus Virus/genital warts:

This is transmitted from person to person through intimate skin contact. It infects mostly the skin and epithelial cells. This is the most common sexually transmitted infection which causes warts on different parts of the body depending on the variety of infection. It transmitted through intimate skin to skin contact. Especially by direct sexual contact such as vaginal, anal, and oral sex. Some types/strains of HPV may not cause any health issues while some can lead to cold sores, genital warts on surrounding skin, inside the mouth or in the throat. even cancers of the cervix, anus and throat.(and penile).The strains that causes various cancer varies.

People infected with HPV are advised to seek medical assistance instantly if you notice any unusual bumps or lesions on the penis, scrotum or anus.

This is a viral infection that causes warts in various parts of the body. It spreads by sexual contact, it's usually a non curable chronic infection which can last for several years or lifelong. Some types of Human Papilloma virus can progress to cervical cancer, anal, mouth and throat cancers e.t.c. Getting vaccinated and receiving regular Pap smears can prevent cervical cancer.

Herpes Infection:

This is a common viral infection that causes painful blisters or ulcers. It is caused by **herpes simplex virus** and it spreads by skin to skin contact, though not curable but it can be treated. There are two types of HSV: Type 1(HSV-1) and Type 2(HSV-2).

TYPE 1(Oral herpes or cold sores)- This is spread by oral contact ,and it causes infection in or around the mouth. Most adults are re-infected by this type of HSV

TYPE 2-(Genital herpes)- This is spread by sexual contact and it causes genital herpes. This infection causes painful blisters or ulcers that can recur over time, medications can reduce the symptoms but can't cure the infections(It is stigmatizing, with impact on sexual relationship), though have mild or no symptoms. Though the recurrent symptoms of both oral and genital herpes may be distressing .

Oral herpes.

Poliomyelitis (Polio):

This is a disease caused by **polio virus**. It causes mild or no symptoms in most people, but it can cause paralysis or death. It is highly infectious viral disease that affects mostly children under the age of 5. There are about three variations of polio virus 1,2 and 3(2 and 3 have been eradicated).This virus is transmitted from person to person through the fecal -oral-route or less frequently by consuming contaminated food or water. This multiplies in the intestine from where it can invade the nervous system and cause paralysis. It's initial symptoms are fever, fatigue, headache, fatigue, vomiting, stiffness of the neck and pain in the limbs. There is no cure currently for polio but can be prevented with polio vaccine given multiple times and can protect the child for a lifetime. There are oral and inactivated polio(intramuscular) vaccines. Those at risk are Live or travel to places where polio is prevalent, poor sanitation areas, children under 5 years of age.

Rabies:

This is a zoonotic, viral disease affecting the central nervous system. Rabies spreads to people and animals via saliva, usually through bites, scratches, or direct contact with mucosa (such as eyes, mouth, open wounds) and it can be fatal. Dogs are responsible for the virus transmission of rabies in human. This infects mammals, dogs, livestock, wildlife. Causative organism of rabies is the rabies virus and other animals that can transmit rabies virus are bats, raccoons, foxes, jackals, cats etc.

Mpox (Formerly called Monkey pox):

This is a viral disease caused by the monkey pox virus(genus Orthopxvirus),it can be transmitted to humans through physical contact, kissing, or sex (vaginal, oral and anal) as well as from respiratory droplets,from an infectious persons, with animal when hunting ,skinning or cooking it, or contaminated materials such as bed sheets, clothes, needles, pregnant mothers may also pass the virus to their unborn babies. It presents symptoms such as severe skin rashes accompanied by fever, headache, muscle aches, back pain, swollen lymph nodes, low energy. Laboratory confirmation is by PCR(testing the skin lesion). Treatment is by supportive care, as well as vaccine help prevent the infection for people at risk.

Zika Virus:

This is a viral disease mostly transmitted through the bite of an infected aedes mosquito, mostly found in the tropics and subtropics. It can also be transmitted from mother to foetus during pregnancy which can lead to certain birth defects, during sexual contact, blood transfusion as well as organ transplants. It's mostly asymptomatic(sometimes present rashes, fever, headache, muscle and joint pain) The main management is on relieving the symptoms and this includes rest, oral rehydration, fever and pain medications such as NSAID's-Non steroidal anti-inflamatory drugs.

Hepatitis:

This refers to an **inflammatory condition of the liver** as a result of viral infection, autoimmune hepatitis(when the body makes antibodies against the liver tissues) as well as hepatitis that occur as a result of medications, drugs(abuse), toxins(exposure), and alcohol which can lead to liver damage as well as inflammation. The five main viral classifications of hepatitis are
Hepatitis A, B, C, D, E, F . Hepatitis A and E causes acute infection, while B, C, and D causes both acute and chronic infections.
The common route of transmission of these viruses are
Hepatitis A-Exposure to Hepatitis A Virus in food or water.
Hepatitis B- contact with Hepatitis B Virus in body fluids, such as blood, vaginal secretions or semen.
Hepatitis C- contact with Hepatitis C Virus in the body fluids(blood, vaginal secretions or semen).
Hepatitis D- Contact with blood Hepatitis D Virus.
Hepatitis E-exposure to Hepatitis E Virus in food or water.